

ECEN 215, Spring 2024

Electrical Circuit Theory

Lecture 13: Step and Natural Response

Ramy Rady

Dept. of Electrical and Computer Engineering

Texas A&M University

ramyrady@tamu.edu



TEXAS A&M
UNIVERSITY

®

OUTLINE



- Quiz1
 - General Solution to transient circuit analysis
 - Natural/Step Response R-C/R-L Circuits
 - Examples
 - Next Lecture
-

OUTLINE

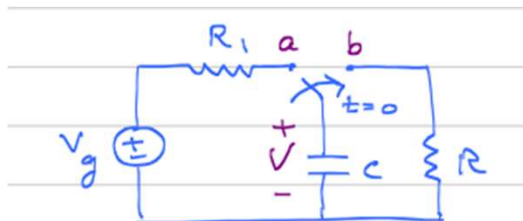


- Quiz1
 - General Solution to transient circuit analysis
 - Natural/Step Response R-C/R-L Circuits
 - Examples
 - Next Lecture
-

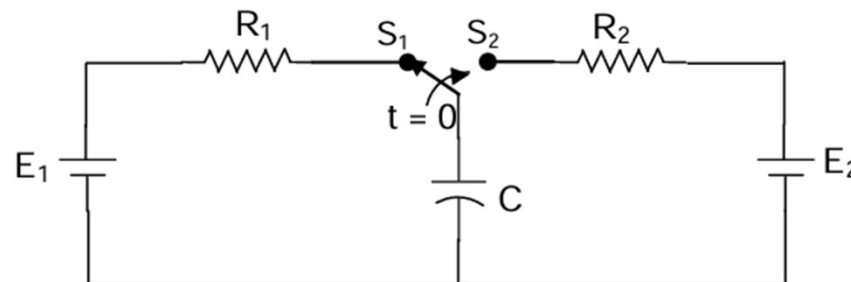
Step Vs. Natural Response



- Natural or source-free circuit
- After switching, is there an independent source ? **No!**
- Still there might be initial charging phase (prior to switching) where there was a source



- Step or driven circuit
- After switching, is there an independent source ? **Yes!**
- Still there might be initial charging phase (prior to switching) where there was a different source





Generic Steps to solve transient circuits

General solution for step and natural responses

For computing the step or natural response of RL and RC circuits follow these steps:

1. Identify the variable of interest. Choose inductor current in an RL and capacitor voltage in an RC circuit.
2. Determine the **initial value** of the variable using the circuit before switching when all inductors are short and capacitors are open at steady state.
3. Determine **the final value** of the variable using the circuit after switching when all inductors are short and capacitors are open at steady state.
4. Calculate the **time constant** by finding the equivalent resistance connected to the inductor or capacitor in the circuit after switching. ($\tau = L/R$ in an RL circuit and $\tau = RC$ in an RC circuit)
5. Use this general equation to find the expression for the variable:

$$X(t) = X_f + [X_0 - X_f] e^{-t/\tau}$$

$X(t)$: unknown variable (i_L or V_c)

X_f : final value, τ : time constant ($\frac{L}{R}$ or RC)

$X(t_0)$: initial value

Special case

- 1) Natural response ($X_f = 0$)

$$X(t) = X_0 e^{-t/\tau}$$

- 2) Step response with no initial charge

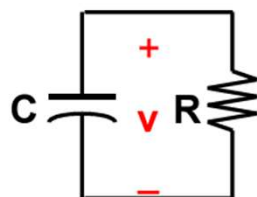
$$X(t) = X_f (1 - e^{-t/\tau})$$

Generic Steps to solve transient circuits (2)

Switching occurs at time $t = 0$

$$v_C(0^+) = v_C(0^-) \quad i_L(0^+) = i_L(0^-)$$

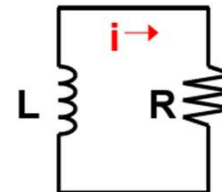
RC Circuit



- Capacitor voltage cannot change instantaneously

$$v(0^-) = v(0^+)$$

RL Circuit



- Inductor current cannot change instantaneously

$$i(0^-) = i(0^+)$$

These circuits are shown for natural response, but you can imagine the same happens when we have step response

With DC excitation, at steady state

capacitor acts as OPEN CIRCUIT and

inductor acts as SHORT CIRCUIT

OUTLINE



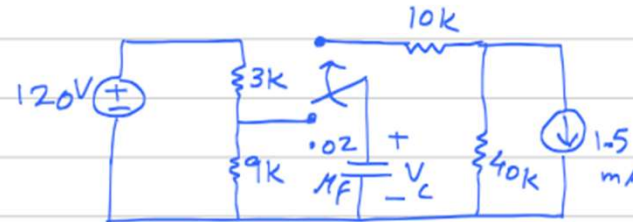
- Quiz1
 - General Solution to transient circuit analysis
 - Natural/Step Response R-C/R-L Circuits
 - Examples
 - Next Lecture
-

Example 1

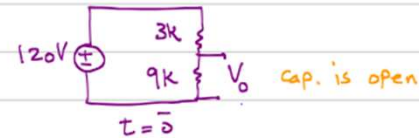


Ex. Find a) $v_c(0)$, b) $v_c(\infty)$, c) τ ,

d) the time required for the capacitor voltage to reach zero after switching.

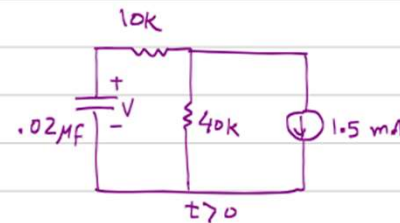


$$V_0 = 120 \frac{9}{9+3} = 90 \text{ V}$$



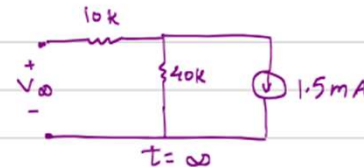
$$\tau = R_{eq} \cdot C = (10k + 40k) \cdot (0.02 \mu F) = 1 \text{ ms}$$

$$V_\infty = -(40k)(1.5 \text{ mA}) = -60 \text{ V}$$



$$V_c(t) = -60 + (90 - (-60)) e^{-\frac{t}{1 \text{ ms}}}$$

$$V_c(t) = -60 + 150 e^{-1000t} \text{ V}$$

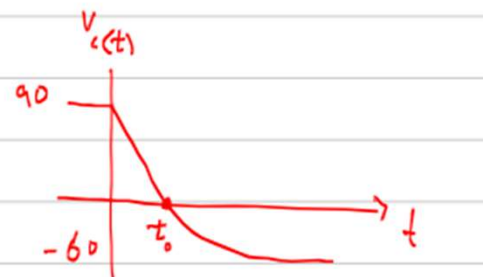


Example 1--Cont.



$$V_C(t) = 0 \rightarrow 150e^{-1000t_0} = 60$$

$$\underline{t_0 = 916.3 \mu\text{s}}$$



Example 2

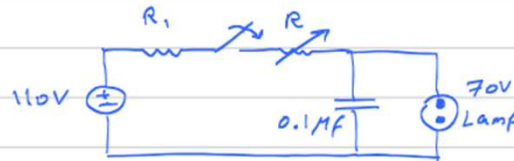


Practical Example, Delay Circuit

This circuit is an RC circuit that provides various time delays.

when the switch is closed, the

capacitor voltage increases toward 110V.



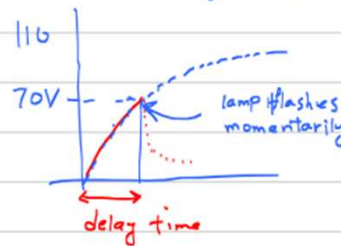
when it reaches 70V, the lamp conducts. Because of the low resistance

of the lamp, the time constant decreases and the capacitor voltage drops

fast and the lamp goes off.

$$\tau_{\text{charge}} = (R + R_1) C, \quad (\text{Lamp off})$$

$$\tau_{\text{discharge}} = \left[(R_1 + R) \parallel R_{\text{lamp}} \right] C, \quad (\text{Lamp on})$$



by adjusting R , the time constant is changed

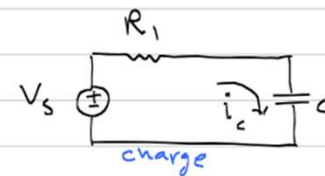
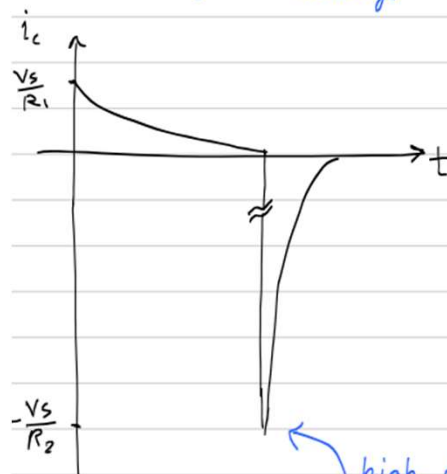
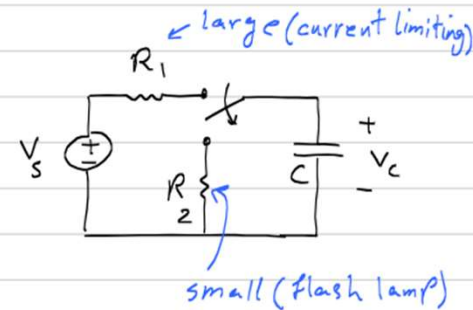
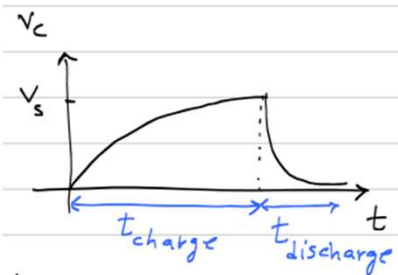
and the delay time before the lamp flashes,

is changed.

Example 2



Practical Example, Photo Flash



high discharge current with the peak of $\frac{V_s}{R_2}$ for a short duration.

Example 2



If $V_S = 80 \text{ V}$, $C = 2 \text{ mf}$, $R_1 = 2 \text{ k}\Omega$, $R_2 = 100 \Omega$, find charge time, discharge time and peak flash current.

$$t_{\text{charge}} = 5\tau_1 = 5R_1C = 5(2000)(2 \times 10^{-3}) = \underline{20 \text{ sec.}}$$

$$t_{\text{discharge}} = 5\tau_2 = 5R_2C = 5(100)(2 \times 10^{-3}) = \underline{1 \text{ sec.}}$$

$$i_{\text{CPK}} = \frac{V_S}{R_2} = \frac{80}{100} = \underline{800 \text{ mA}}$$

Next Lecture

- Phasors → After Spring break

